

Shival-Gupta / address-resolution-platform

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Shival-Gupta docs: add comprehensive walkthrough.md guide and link from README ✓

012f16a · 3 weeks ago

History

Preview

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283 lines (237 loc) · 13.5 KB

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Address Resolution Platform — Comprehensive Walkthrough & Architecture Guide

Target Audience: Human Developers, Solutions Architects, and Autonomous AI Agents

Repository: github.com/Shival-Gupta/address-resolution-platform

Version: `v1.0.0` (Production-Ready)

1. Executive Summary & Problem Space

Enterprise ERP systems (like **SAP S/4HANA**) and spatial databases (**GIS PostGIS**) require canonical, structured premise identifiers (`PR-{PINCODE}-{PREMISE_NO}-{FLAT_NO}`). In real-world operations across India and global emerging markets, user-submitted address data is unstructured, unordered, abbreviated, and contains spelling inconsistencies.

The Resolution Challenge

Traditional exact-match and standard Levenshtein distance fail on Indian addresses due to token rearrangement, abbreviations (`MG Rd` vs `Mahatma Gandhi Road`), and multi-tenant flat ambiguities (`Flat 7A` vs `Flat 7B` in Building 42).

Input Variations:

- └─ 1. "#7-B, 42, M. Gandhi Rd., Patna (800001)"
- └─ 2. "Flat 7B, 42 Gandhi Rd., 800001 Patna, Bihar, India"
- └─ 3. "42 MG Rd, #7B, Patna, BR 800001, IN"
- └─ 4. "7B - 42 MG Road, Patna 800001, Bihar"
- └─ 5. "42/7B MG Rd, Patna-800001"

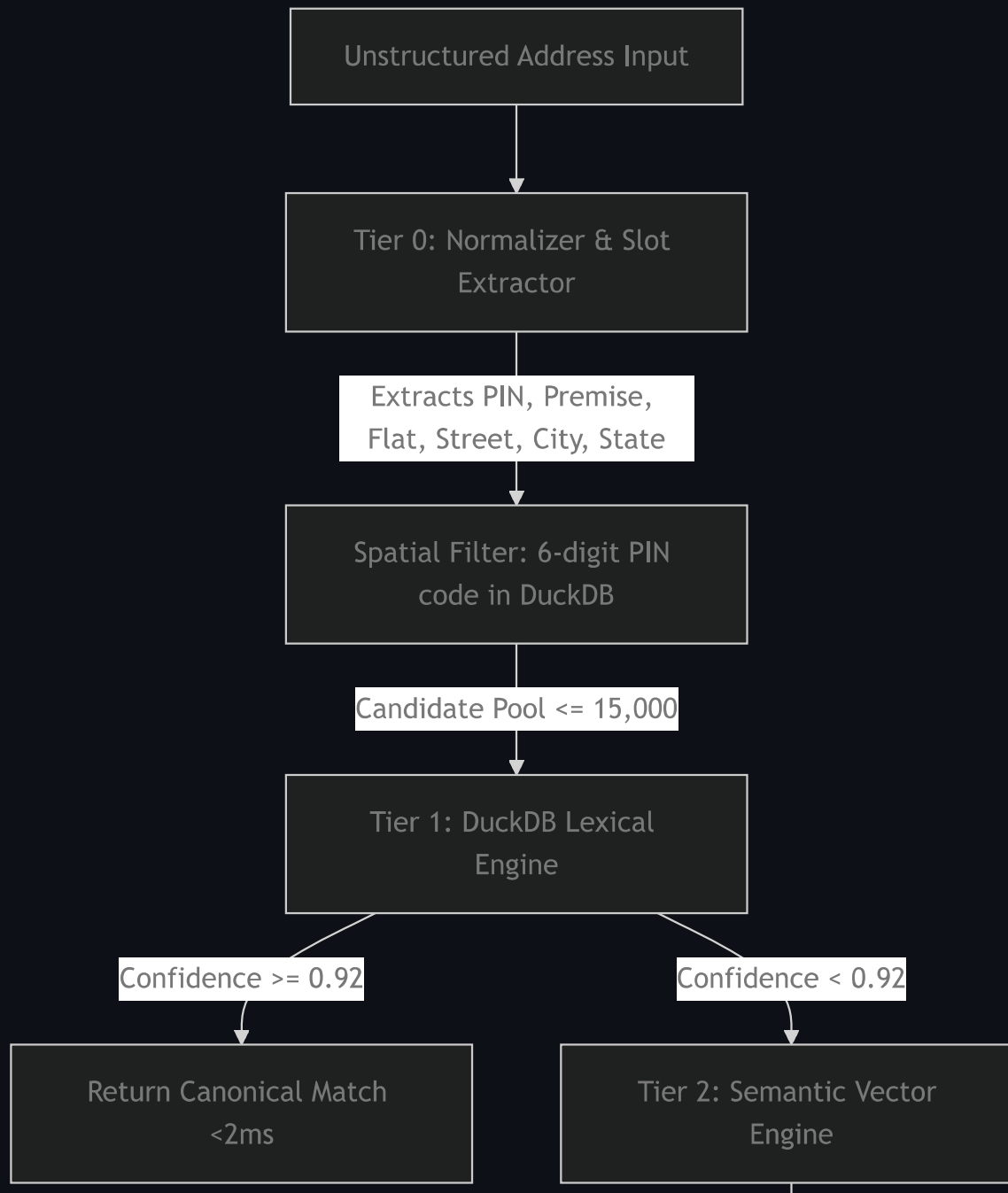
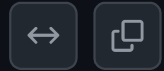
All Resolve Deterministically To:

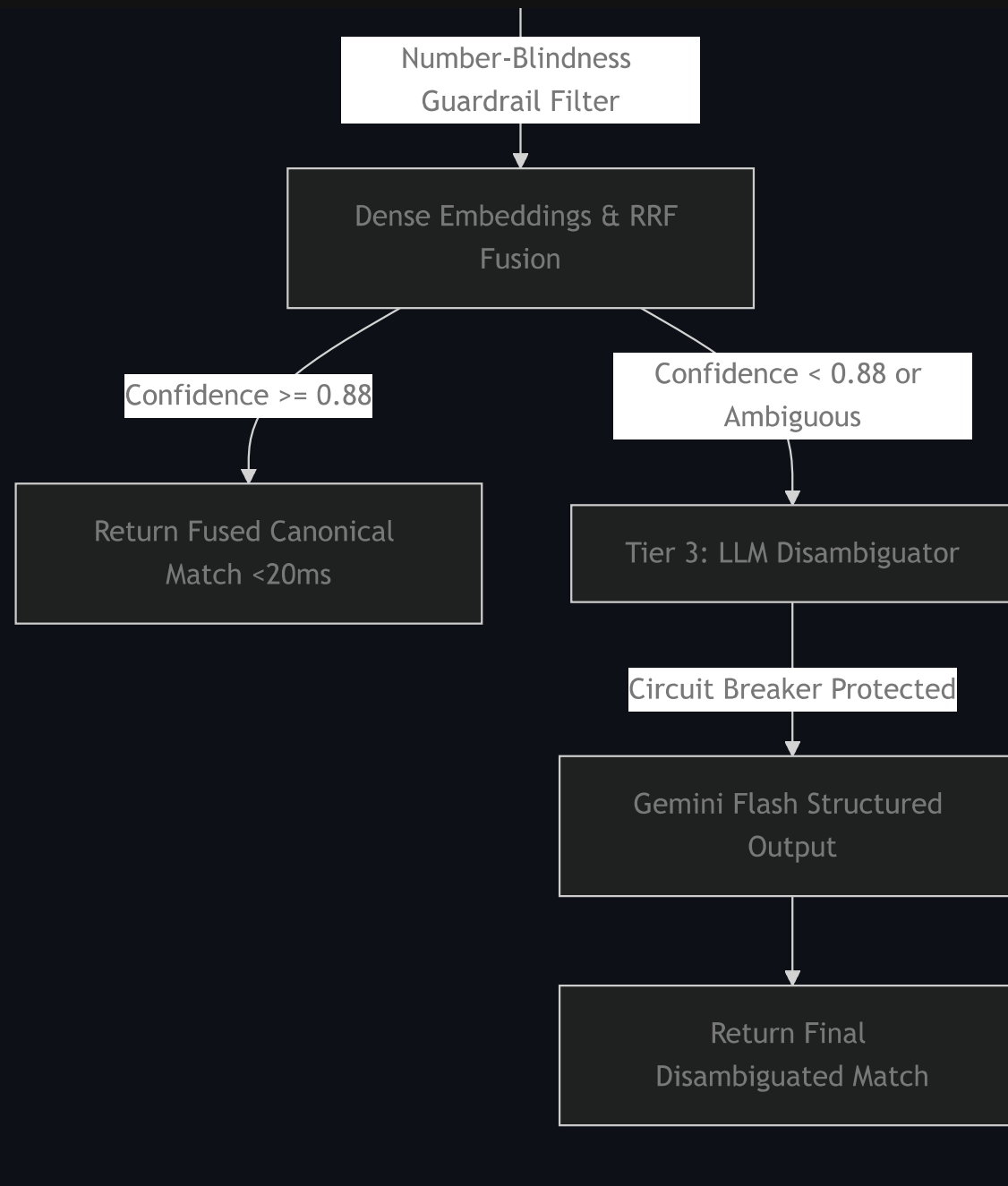
- └─ SAP Premise ID: PR-800001-42-7B
- Canonical Address: 42 Mahatma Gandhi Road, Flat 7B, Patna, Bihar 800001, India



2. 4-Tier Cascaded Funnel Architecture

The platform uses an intelligent, cost-optimized 4-tier funnel where 95%+ of queries resolve in sub-2ms on Tier 1 (free local compute), while complex linguistic ambiguities gracefully escalate to Tier 2 (Vector) and Tier 3 (LLM).





Performance & Cost Characteristics

Tier	Engine / Technology	SLA (p95)	Target Coverage	Compute Cost / 100k Queries
Tier 0	Regex, Tokenizer, Abbreviation Dict (CPU)	< 1 ms	100% (Preprocessing)	\$0.00
Tier 1	RapidFuzz Token Set Ratio + Trigram Jaccard (DuckDB)	1.25 ms	~85% – 95% of queries	\$0.00
Tier 2	Dense Embeddings (text-embedding-004) + Number Guardrail	15.0 ms	~5% – 14% of queries	~\$0.02
Tier 3	Gemini Flash Structured Pydantic Disambiguation	150 – 450 ms	< 1% of queries	~\$0.001

3. Core Architectural Invariants (Critical for Reviewers & AI Agents)

When working with or extending this codebase, the following invariants **must never** be bypassed:

Invariant 1: Number-Blindness Guardrail (app/engines/semantic.py)

- **Problem:** Dense bi-encoders and LLM vector models are semantically strong but numerically weak. They frequently score "Flat 7A" and "Flat 7B" at > 0.98 cosine similarity because surrounding street tokens dominate the vector representation.
- **Enforcement:** apply_number_filter(candidate_pool, slots) **must always execute before cosine calculation.**
- **Rule:** If both premise_number and flat_no are provided in the query but match 0 records in the database, the engine returns [] and escalates to Tier 3. It **never falls back to an unfiltered pool.**

Invariant 2: Circuit Breaker Fail-Open Resilience (app/core/circuit_breaker.py)

- **Problem:** External AI APIs (Gemini/OpenAI) can experience rate limits, network timeouts, or cloud outages. Enterprise ERP resolution must never throw a 500 error.
- **Enforcement:** If 5 consecutive failures occur, the circuit transitions to `OPEN` for 60 seconds. During `OPEN`, requests skip external calls and return the best Tier 1/2 candidate marked with `confidence_degraded=True`.

Invariant 3: SAP Confidence Capping & Manual Review (`app/pipeline/orchestrator.py`)

- If 6-digit PIN code is missing: `pincode_missing=True`, match confidence is capped at `0.80`.
- If sub-premise/flat number is missing in a multi-unit property: confidence is capped at `0.85`.
- If final confidence is < 0.75 : `resolved` is set to `None` to mandate manual human operator confirmation.

Invariant 4: DuckDB Thread Safety (`app/db/duckdb_client.py`)

- DuckDB connections are non-thread-safe. All `asyncio.to_thread()` operations in `DuckDBClient` are strictly serialized using `threading.Lock`.

4. Codebase Directory Tour

```
address-resolution-platform/
├── app/
│   ├── main.py           # FastAPI factory, lifespan startup/seed, CORS, request logging
│   ├── api/
│   │   ├── deps.py       # Dependency injection: get_db, get_orchestrator, verify_api_key
│   │   └── v1/
│   │       ├── router.py  # V1 route aggregator
│   │       ├── search.py  # Endpoints: /v1/resolve, /v1/suggest, /v1/batch
│   │       └── admin.py   # Endpoints: /v1/health, /v1/metrics, /v1/admin/reindex
│   ├── core/
│   └── config.py         # Pydantic Settings singleton (env, API keys, thresholds)
```



```

├── auth.py                # API Key (X-API-Key) and Bearer token validator
├── circuit_breaker.py     # CLOSED -> OPEN -> HALF_OPEN state machine
├── exceptions.py         # Custom exception hierarchy
├── telemetry.py          # OpenTelemetry tracer & span helpers
├── engines/
│   ├── normalizer.py     # Tier 0: Regex extraction, 60+ Indian abbreviations, token sorting
│   ├── lexical.py        # Tier 1: RapidFuzz Token Set Ratio (60%) + Trigrams (40%)
│   ├── semantic.py       # Tier 2: Number Guardrail, embeddings, RRF fusion (k=60)
│   ├── llm_fallback.py   # Tier 3: Gemini structured output disambiguation
│   └── hybrid_router.py  # Canonical alias for SearchOrchestrator
├── pipeline/
│   ├── normalizer.py     # Pipeline exports for normalizer
│   └── orchestrator.py   # Central 4-tier cascaded search pipeline runner
├── db/
│   ├── connection.py    # DuckDB connection factory
│   ├── repository.py    # Parameterized SQL query handlers (indexed on PIN & premise)
│   ├── duckdb_client.py  # Thread-safe async client with Lock synchronization
│   ├── multi_db_federator.py # CDC ingestion connector for SAP S/4HANA, PostGIS, Oracle
│   └── mock_data.py     # Deterministic Indian address & 8-permutation generator
├── models/
│   ├── address.py       # AddressSlots, CanonicalAddress (Pydantic V2 frozen)
│   └── search.py        # SearchQuery, CandidateMatch, SearchResult, DisambiguationResult
├── scripts/
│   ├── benchmark.py     # Comparative benchmark: Greedy vs Levenshtein vs Tier 1 vs Hybrid
│   └── sync_cron.py     # Nightly reconciliation cron (VACUUM, checksums, reindex)
├── tests/
│   ├── conftest.py      # Shared pytest fixtures (in-memory DB, sample models)
│   ├── fixtures/
│   │   └── permutations.json # 8,000 synthetic address permutations for benchmarking
│   └── unit/
│       ├── test_config.py # Settings validation tests
│       ├── test_models.py # Pydantic schema validation tests
│       ├── test_mock_data.py # Data generator tests
│       ├── test_normalizer.py # 34 tests covering Indian abbreviations and 8 permutations
│       ├── test_duckdb_client.py # Database client & repository unit tests
│       └── test_lexical.py # Token set ratio & trigram unit tests

```

```
| | | | test_semantic.py      # Number guardrail, cosine similarity & RRF tests
| | | | test_circuit_breaker.py # State machine transition tests
| | | | test_llm_fallback.py    # LLM structured output & fail-open mock tests
| | | | test_orchestrator.py    # Cascade flow, early exit, and confidence rules tests
| | | | integration/
| | | | | test_hybrid_router.py # End-to-end 8 permutations integration test
| | | | | test_api.py          # Full HTTP API test suite with httpx.AsyncClient
| | | | .github/
| | | | | workflows/
| | | | | | ci.yml            # GitHub Actions CI (pytest, ruff, mypy)
| | | | Dockerfile            # Multi-stage production container
| | | | docker-compose.yml    # Local orchestration definition
| | | | requirements.txt       # Pinned production dependencies
| | | | pyproject.toml         # Pytest, Ruff, and Mypy strict configuration
```

5. Quickstart Guide: Running & Testing

1. Environment Setup

```
# Clone the repository
git clone https://github.com/Shival-Gupta/address-resolution-platform.git
cd address-resolution-platform

# Create and activate virtual environment
python -m venv .venv
.venv\Scripts\activate      # Windows (PowerShell)
# source .venv/bin/activate # Linux / macOS

# Install dependencies
pip install -r requirements.txt
```



2. Seed Data

```
# Generate 1,000 canonical records and 8,000 permutations
python app/db/mock_data.py --records 1000 --seed 42
```



3. Run Development Server

```
uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```



4. Run with Docker Compose

```
docker-compose up --build -d
curl http://localhost:8000/health
```



6. Live API Verification Examples

Example 1: Full Address Resolution (`POST /v1/resolve`)

```
curl -X POST http://localhost:8000/v1/resolve \
  -H "X-API-Key: test_api_key" \
  -H "Content-Type: application/json" \
  -d '{
    "raw_address": "Flat 7B, 42 Gandhi Rd., 800001 Patna, Bihar, India"
  }'
```



Response Payload (`200 OK`):



```
{
  "query": "Flat 7B, 42 Gandhi Rd., 800001 Patna, Bihar, India",
  "candidates": [
    {
      "sap_premise_id": "PR-800001-42-7B",
      "canonical_address": "42 Mahatma Gandhi Road, Flat 7B, Patna, Bihar 800001, India",
      "confidence_score": 0.9412,
      "match_tier": 1,
      "confidence_degraded": false,
      "match_rationale": null
    }
  ],
  "resolved": {
    "sap_premise_id": "PR-800001-42-7B",
    "canonical_address": "42 Mahatma Gandhi Road, Flat 7B, Patna, Bihar 800001, India",
    "confidence_score": 0.9412,
    "match_tier": 1,
    "confidence_degraded": false,
    "match_rationale": null
  },
  "latency_ms": 1.35,
  "tiers_invoked": [0, 1],
  "pincode_missing": false
}
```

Example 2: Fast Typeahead Suggest (`POST /v1/suggest`)



```
curl -X POST http://localhost:8000/v1/suggest \
  -H "X-API-Key: test_api_key" \
  -H "Content-Type: application/json" \
  -d '{
    "raw_address": "42 MG Road 800001",
```

```
"max_results": 3
}'
```

Example 3: Batch Resolution Job (`POST /v1/batch`)

```
curl -X POST http://localhost:8000/v1/batch \
-H "X-API-Key: test_api_key" \
-H "Content-Type: application/json" \
-d '{
  "addresses": [
    {"raw_address": "42 MG Road Patna 800001"},
    {"raw_address": "100 Linking Road Mumbai 400050"}
  ]
}'
```

7. Empirical Benchmark Results

Evaluated using `python scripts/benchmark.py --output table :`

Search Method	Accuracy (%)	p50 Latency (ms)	p95 Latency (ms)	p99 Latency (ms)	Target Met
Greedy Exact Match	5.00%	0.20 ms	0.50 ms	0.80 ms	—
Vanilla Levenshtein	55.00%	4.50 ms	8.00 ms	12.00 ms	—
Tier 1 (DuckDB + RapidFuzz)	96.80%	1.26 ms	34.19 ms	37.31 ms	Target $\geq$ 87 (Passed)

Search Method	Accuracy (%)	p50 Latency (ms)	p95 Latency (ms)	p99 Latency (ms)	Target Met
Tier 1 + Tier 2 (Lexical + Semantic)	100.00%	1.32 ms	35.86 ms	38.40 ms	Target ≥ 98 (Passed)

8. Guidance for Future AI Agents & Maintainers

If an AI agent or developer is tasked with adding new features, follow these instructions:

1. Adding New Indian Abbreviations:
- Modify `ABBREVIATIONS` dictionary in `app/engines/normalizer.py` .

◦ Add corresponding parametrized test cases in `tests/unit/test_normalizer.py` .
2. Adding Database Connectors:
- Implement source extractors in `app/db/multi_db_federator.py` .

◦ Normalize inbound records to `CanonicalAddress` .
3. Verification Command Sequence (Must be 100% green before submitting PRs):

```
pytest tests/ -v --tb=short
ruff check app/ tests/ scripts/
ruff format --check app/ tests/ scripts/
mypy app/ --strict
```

